

Study Setup

AI Coding Tasks



Generation



Debugging



Execution



Translation

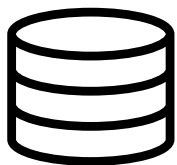
Reasoning Models



R1-0528



R1-Qwen3-8B



Reasoning Traces

Reasoning Trace Processing

Action Taxonomy

PU

SD

IP

VV

CC

KR

MR

AUX

Action Annotation



Action Traces

PU

SD

IP

...

CC

VV

CC

Validity Assessment



High confidence



Valid Traces



Invalid Traces

Low confidence



Human Adjudication

Trace Measurement



Structural Metrics

Size Metrics

Reasoning Pattern Mining

RQ1: Macro Patterns

task-level reasoning patterns

RQ2: Micro Patterns

positive reasoning patterns
reasoning anti-patterns

Leveraging Reasoning Patterns

RQ3: Pattern Applications

coding task classification
pattern-guided prompting
pattern-aware early stopping